# YONEDA STUFF

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## 1. The Yoneda lemma

### 1.1. Some definitions and notation

Given a (small) category $\mathcal{C}$, we let $\mathcal{P}(\mathcal{C})$ denote the category of *presheaves* on $\mathcal{C}$, defined by

$$\mathcal{P}(\mathcal{C}) := \mathsf{Fun}(\mathcal{C}^{\mathsf{op}}, \mathsf{Set}).$$

In other words, the objects of $\mathcal{P}$ are functors $\mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$ (sometimes called *presheaves on* $\mathcal{C}$), and the morphisms are natural transformations. That is, the set of natural transformations between two such functors $F, G \in \mathcal{P}(\mathcal{C})$ is exactly

$$\mathrm{Hom}_{\mathcal{P}(\mathcal{C})}(F, G).$$

(Sometimes this set might instead be written as something like $\mathsf{Nat}(F, G)$, but this kind of loses the emphasis that this is really the set of maps between to objects in a category.)

We say presheaf $G \in \mathcal{P}(\mathcal{C})$ is *representable* if it is of the form

$$G = \mathrm{Hom}_{\mathcal{C}}(-, c)$$

for some object $c$ of $\mathcal{C}$. (If we're being really precise, we might instead ask that $G$ is *isomorphic* to some $\mathrm{Hom}_{\mathcal{C}}(-, c)$ as an object of $\mathcal{P}(\mathcal{C})$ instead of *equal*.)

Given a morphism $f : c \to d$ in $\mathcal{C}$, there are natural transformations

$$\mathrm{Hom}_{\mathcal{C}}(-, c) \xrightarrow{f_*} \mathrm{Hom}_{\mathcal{C}}(-, d),$$
$$\mathrm{Hom}_{\mathcal{C}}(d, -) \xrightarrow{f^*} \mathrm{Hom}_{\mathcal{C}}(c, -).$$

Concretely, given a map $g \in \mathrm{Hom}_{\mathcal{C}}(b, c)$, we have $f_*(g) = f \circ g \in \mathrm{Hom}_{\mathcal{C}}(b, d)$, and given a map $h \in \mathrm{Hom}_{\mathcal{C}}(d, e)$, we have $f^*(h) = h \circ f \in \mathrm{Hom}_{\mathcal{C}}(c, e)$. We can keep track of this in the following commutative diagram:

$$\begin{tikzcd}
b \arrow[r, "g"] \arrow[rr, bend right, "f_*g"'] & c \arrow[r, "f"] \arrow[rr, bend left, "f^*h"] & d \arrow[r, "h"] & e
\end{tikzcd}$$

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**Remark 1.1.1.** Recall that the a *natural transformation* $\eta$ between two functors $F, G : \mathcal{C} \to \mathcal{D}$ is an assignment, for every morphism $f : c \to c'$ in $\mathcal{C}$, two morphisms $\eta_c : F(c) \to G(c)$ and $\eta_{c'} : F(c') \to G(c')$ in $\mathcal{D}$ which make the following diagram commutes in $\mathcal{D}$:

$$\begin{tikzcd} F(c) \arrow[r, "F(f)"] \arrow[d, "\eta_c"'] & F(d) \arrow[d, "\eta_d"] \\ G(c) \arrow[r, "G(f)"] & G(d). \end{tikzcd}$$

A special example for us is if $F$ and $G$ are functors in $\mathcal{P}(\mathcal{C})$ and $F = \mathrm{Hom}_{\mathcal{C}}(-, c)$, in which case a morphism $g : a \to b$ gives us a diagram

$$\begin{tikzcd} \mathrm{Hom}_{\mathcal{C}}(b, c) \arrow[r, "g^*"] \arrow[d, "\eta_b"'] & \mathrm{Hom}_{\mathcal{C}}(a, c) \arrow[d, "\eta_a"] \\ F(b) \arrow[r, "F(g)"] & F(a). \end{tikzcd}$$

This diagram will show up in the proof of the Yoneda lemma.

## 1.2. **The lemma**

**Theorem 1.2.1** (Yoneda Lemma)**.** *Let $F \in \mathcal{P}(\mathcal{C})$, and let $c$ be an object of $\mathcal{C}$. There is a bijection of sets*

$$\mathrm{Hom}_{\mathcal{P}(\mathcal{C})}(\mathrm{Hom}_{\mathcal{C}}(-, c), F) \cong F(c).$$

*Moreover this bijection is* natural*, i.e., for every map $f : c \to d$ in $\mathcal{C}$, we get a commuting square*

$$\begin{tikzcd} \mathrm{Hom}_{\mathcal{P}(\mathcal{C})}(\mathrm{Hom}_{\mathcal{C}}(-, d), F) \arrow[r, "\cong"] \arrow[d] & F(d) \arrow[d, "F(f)"] \\ \mathrm{Hom}_{\mathcal{P}(\mathcal{C})}(\mathrm{Hom}_{\mathcal{C}}(-, c), F) \arrow[r, "\cong"] & F(c) \end{tikzcd}$$

**Remark 1.2.2.** The left-hand vertical arrow in the diagram above is the map of sets which sends a natural transformation $\eta : \mathrm{Hom}_{\mathcal{C}}(-, d) \to F$ to the composition of natural transformations

$$\mathrm{Hom}_{\mathcal{C}}(-, c) \xrightarrow{f_*} \mathrm{Hom}_{\mathcal{C}}(-, d) \xrightarrow{\eta} F$$

**Remark 1.2.3.** There's also a "covariant version" of the Yoneda lemma which we can get by just taking $\mathcal{D} = \mathcal{C}^{\mathsf{op}}$ in the above theorem. It would look something like this. Given a functor $F : \mathcal{D} \to \mathsf{Set}$ and object $d \in \mathcal{D}$, we have

$$\mathrm{Hom}_{\mathcal{P}(\mathcal{C})}(\mathrm{Hom}_{\mathcal{D}}(d, -), F) \cong F(d).$$

*Proof of Theorem 1.2.1.* We'll construct a bijection of sets

$$A : \mathrm{Hom}_{\mathcal{P}(\mathcal{C})}(\mathrm{Hom}_{\mathcal{C}}(-, c), F) \cong F(c),$$

but leave out the part about checking that $A$ is natural. Let $\eta$ be a natural transformation from $\mathrm{Hom}_{\mathcal{C}}(-, c)$ to $F$. As part of the data of a natural transformation, we get a map of sets

$$\eta_c : \mathrm{Hom}_{\mathcal{C}}(c, c) \to F(c).$$

The set $\mathrm{Hom}_{\mathcal{C}}(c, c)$ always has an identity element $1_c$. We'll use this to define $A(\eta) = \eta_c(1_c)$ as an element of $A(c)$. For the rest of the proof, we'll check that this $A$ is both injective and surjective.

To see that $A$ is injective, we check that whenever $\eta$ and $\eta'$ are natural transformations from $\mathrm{Hom}_{\mathcal{C}}(-, c)$ to $F$ such that $A(\eta) = A(\eta')$, then in fact $\eta = \eta'$. For $\eta$ and $\eta'$ to be equal, we need $\eta_d(f) = \eta'_d(f)$ for every object $d \in \mathcal{C}$ and morphism $f : d \to c$. Since $\eta$ is a natural transformation, we have a commutative square (of sets and maps of sets)

$$\begin{tikzcd} \mathrm{Hom}_{\mathcal{C}}(c, c) \arrow[r, "f^*"] \arrow[d, "\eta_c"'] & \mathrm{Hom}_{\mathcal{C}}(d, c) \arrow[d, "\eta_d"] \\ F(c) \arrow[r, "F(f)"] & F(d) \end{tikzcd} \tag{1.2.4}$$

Consider the element $1_c$ in $\mathrm{Hom}_{\mathcal{C}}(c,c)$. Following the left and bottom composition in this square, we find that $1_c$ is sent to $A(\eta)$, and then $F(f)(A(\eta))$. On the other hand, following the top and right compositions, we find $1_c$ is sent to $f$, and then to $\eta_d(f)$. Thus,

$$F(f)(A(\eta)) = \eta_d(f).$$

But an identical argument can be made for the natural transformation $\eta'$, and so we find that

$$\eta_d(f) = F(f)(A(\eta)) = F(f)(A(\eta')) = \eta'_d(f),$$

as desired.

On the other hand, we see that $A$ is surjective, since given any element $x \in F(c)$ we can define a natrual transformation $\eta$ from $\mathrm{Hom}_{\mathcal{C}}(-,c)$ to $F$ via the diagram (1.2.4) by

$$\eta_d(f) := F(f)(x).$$

By construction, this $\eta$ satisfies

$$A(\eta) = \eta_c(1_c) = F(1_c)(x) = 1_{F(c)}(x) = x. \qquad \square$$

The following is a corollary of the above formulation of the Yoneda lemma. It tends to be somewhat more useful, and often people will just call this the Yoneda lemma instead.

**Corollary 1.2.5** (Yoneda embedding)**.** *The functor*

$$y : \mathcal{C} \to \mathcal{P}(\mathcal{C}),$$

*defined on objects by* $y(c) = \mathrm{Hom}_{\mathcal{C}}(-,c)$ *and on morphisms by* $y(f) = f_*$*, is fully faithful.*

*Proof.* By definition, to show that $y$ is fully faithful, we need to show that the map of sets

$$\mathrm{Hom}_{\mathcal{C}}(c,d) \xrightarrow{y} \mathrm{Hom}_{\mathcal{P}(\mathcal{C})}(y(c), y(d))$$

is a bijection for any choice of $c, d \in \mathcal{C}$. This is exactly the statement of Theorem 1.2.1 when we take

$$F = y(d) = \mathrm{Hom}_{\mathcal{C}}(-,d). \qquad \square$$

The Yoneda embedding lets us think of $\mathcal{C}$ as a full subcategory of the much larger, but generally nicer category $\mathcal{P}(\mathcal{C})$. For example, $\mathcal{P}(\mathcal{C})$ admits all colimits. In fact, in a precise sense, it's "generated" by the image of $y$ via colimits. This is a fancy way of saying that every object of $\mathcal{P}(\mathcal{C})$, representable or not, is "built from" objects in $\mathcal{C}$.

## 2. Universal properties via the Yoneda lemma

### 2.1. What even are universal properties?

One central philosophy in category theory is that you should think more about morphisms between objects, rather than the objects themselves. (Haynes Miller calls this attitude "Wittgensteinian" in [Mil21, §39], which is kinda fun.) And so following this philosophy, something we might want to do is to make all our definitions in terms of morphisms.

For example, say we have an object $c$ of a category $\mathcal{C}$, and we want to do some kind of "construction" with it to obtain an object of a (maybe different) category $\mathcal{D}$, i.e., we want to define a functor $F : \mathcal{C} \to \mathcal{D}$. Sometimes, if the categories $\mathcal{C}$ and $\mathcal{D}$ are nice, we can just do this at the level of sets and stuff, like defining the "free group" functor

$$\mathsf{Free} : \mathsf{Set} \to \mathsf{Grp}$$

by saying something like "send a set $X$ to the group of all strings of elements of $X$, and let the group operation be string concatenation." But this isn't always possible (as we will see later), and so we want something more general:

**Goal 2.1.1.** We want to be able to define the functor $F : \mathcal{C} \to \mathcal{D}$ on each object $c$ just by specifying what the set $\mathrm{Hom}_{\mathcal{D}}(d, F(c))$ should be for each object $d$ of $\mathcal{D}$.

But this goal can be rephrased as "we want to define $F$ by specifying a functor $\tilde{F} : \mathcal{C} \to \mathcal{P}(\mathcal{D})$ which takes values in the representable presheaves." That is, we want a correspondence

$$\left\{ \begin{array}{c} \text{Functors} \\ F : \mathcal{C} \to \mathcal{D} \end{array} \right\} = \left\{ \begin{array}{c} \text{Diagrams} \\ \begin{tikzcd} & \mathcal{P}_{\mathrm{rep}}(\mathcal{D}) \arrow[d, hook] \\ \mathcal{C} \arrow[ur] \arrow[r, "\tilde{F}"'] & \mathcal{P}(\mathcal{D}) \end{tikzcd} \end{array} \right\}$$

where $\mathcal{P}_{\mathrm{rep}}(\mathcal{D})$ is the full subcategory of $\mathcal{P}(\mathcal{D})$ consisting of the representable presheaves. But Corollary 1.2.5 tells us that $\mathcal{D} \simeq \mathcal{P}_{\mathrm{rep}}(\mathcal{D})$, so this correspondence works. We give a (rough) definition.

**Definition 2.1.2.** Let $c$ be an object of $\mathcal{C}$, and $F : \mathcal{C} \to \mathcal{D}$. To give a *universal property* of $F(c)$ is to give a characterization of the representable presheaf $y(F(c)) = \mathrm{Hom}_{\mathcal{D}}(-, F(c))$.

Usually the idea of "universal property" is understood through examples. Indeed, this definition is extremely terse and maybe unenlightening, but it's nice to have an explicit foundation for what we mean by "universal property." We'll give some examples below.

**Remark 2.1.3.** Following Remark 1.2.3, we can just as well talk about "universal properties" as characterizations of the "corepresentable" functors $\mathrm{Hom}_{\mathcal{D}}(F(c), -)$, i.e., describing the object $F(c)$ by saying what all the maps *out of it* are, instead of all the maps *into it*. We'll use these two notions interchangeably in the examples below.

**Example 2.1.4** (Universal property of the free group)**.** The *free group* construction is a functor $\mathsf{Free} : \mathsf{Set} \to \mathsf{Grp}$ with the property that for a set $X$, the set of group homomorphisms $\mathrm{Hom}_{\mathsf{Grp}}(\mathsf{Free}(X), G)$ to any group $G$ are in bijection with the set of maps of *sets* from $X$ to the underlying set of $G$, which we denote $\mathsf{Forget}(G)$. For example,

$$\mathrm{Hom}_{\mathsf{Grp}}(\mathsf{Free}(\{a, b\}), G) \cong \mathrm{Hom}_{\mathsf{Set}}(\{a, b\}, \mathsf{Forget}(G)).$$

(This is exactly saying that we have an adjunction $\mathsf{Free} \dashv \mathsf{Forget}$.)

**Example 2.1.5** (Universal property of the quotient of groups)**.** For a group $G$ with normal subgroup $N \leq G$, the *quotient* $G/N$ is characterized by the universal property that set of maps $\mathrm{Hom}_{\mathsf{Grp}}(G/N, H)$ to any other abelian group $H$ are in bijection with the set of maps $\varphi : G \to H$ such that the restriction $\varphi|_N$ is the trivial group homomorphism (i.e., it factors through the unique map $N \to 1$). That is, the set of maps $\varphi$ which fit into the following diagram:

$$\begin{tikzcd} N \arrow[r, hook] \arrow[d] & G \arrow[d, "\varphi"] \\ 1 \arrow[r] & H \end{tikzcd}$$

**Example 2.1.6** (Universal property of the fraction field of a ring)**.** Let $R$ be a ring (commutative and with no zero divisors). The *fraction field* $\mathsf{Frac}(R)$ is the field characterized by the universal property that maps from $\mathsf{Frac}(R)$ to any other field $k$ are in bijection with the set of *ring* homomorphisms from $R$ to $k$:

$$\mathrm{Hom}_{\mathsf{Fields}}(\mathsf{Frac}(R), k) \cong \mathrm{Hom}_{\mathsf{Ring}}(R, k).$$

(This is exactly saying that $\mathsf{Frac}$ is left adjoint to the forgetful functor $\mathsf{Fields} \to \mathsf{Ring}$.) The most important example is

$$\mathsf{Frac}\,(\mathbb{Z}) = \mathbb{Q}\,.$$

Another example is that the fraction field of the ring $\mathbb{R}[x]$ of polynomials over $\mathbb{R}$ is $\mathbb{R}(x)$, the ring of rational functions.

In fact, $R$ is always a subring of $\mathsf{Frac}(R)$, and something fancy we can say is that every map from $R$ to a field $k$ *factors uniquely through* this inclusion. That is, given any map $\varphi : R \to k$, we always have a diagram:

$$\begin{tikzcd} R \arrow[r, "\varphi"] \arrow[d, hook] & k \\ \mathsf{Frac}(R) \arrow[ur, dashed] & \end{tikzcd}$$

This is saying that the inclusion $R \hookrightarrow \mathsf{Frac}\,(R)$ is *initial* among maps from $R$ to a field.

**Example 2.1.7** (Universal property of the tensor product of vector spaces)**.** Given two vector spaces $U, V$ (over $\mathbb{R}$, say), the *tensor product* $U \otimes V$ is the vector space characterized by the universal property that linear maps $U \otimes V \to W$ are in bijection with the set of *bilinear* maps $U \times V \to W$.

In fact, there is a natural linear map $U \times V \to U \otimes V$ such that for any bilinear map $\varphi : U \times V \to W$, we get a unique linear map $\psi : U \otimes V \to W$ making the following diagram commute (in the category of *sets*):

$$\begin{tikzcd} U \times V \arrow[r] \arrow[dr, "\varphi"'] & U \otimes V \arrow[d, dashed, "\psi"] \\ & W. \end{tikzcd}$$

## 2.2. **Universal property of limit and colimit functors**

"Taking the colimit" and "taking the limit" are things we can do to *diagrams* in categories (provided the colimit and limit exist, which they might not). By diagram, we mean a collection of objects and arrows sitting in that category. Roughly, the colimit of a diagram is supposed to be the "initial" object receiving (compatible) maps from all the objects in the diagram, and the limit is the "final" object with (compatible) maps to all the objects. We'll be more precise in a moment, but first some examples.

**Example 2.2.1** (Quotient of groups as a colimit)**.** Let $G$ be a group and $N$ a normal subgroup. One construction of the *quotient* $G/N$ is as the set of "cosets":

$$G/N = \{gN : g \in G\}$$

with the group multiplication $gN \cdot g'N = (gg')N$ and unit $1N$. But, if we squint at the diagram in Example 2.1.5, we can see that the map $G \to G/N$ is "initial" among maps $G \to H$ which restrict to the trivial homomorphism on $N$. That is, for any such $\varphi : G \to H$, we get a map $\psi : G/N \to H$ making the following diagram commute:

$$\begin{tikzcd} N \arrow[r, hook] \arrow[d] & G \arrow[d] \arrow[ddr, bend left, "\varphi"] & \\ 1 \arrow[r] \arrow[drr, bend right] & G/N \arrow[dr, dashed, "\psi"] & \\ & & H \end{tikzcd}$$

In other words, $G/N$ is the *colimit* of the diagram

$$\begin{tikzcd} N \arrow[r, hook] \arrow[d] & G \\ 1. & \end{tikzcd} \tag{2.2.2}$$

**Remark 2.2.3.** Note that in this example, the condition that $N$ be normal is important. If $N$ is not normal, then the quotient $G/N$ doesn't exist, i.e., the diagram (2.2.2) "does not admit a colimit."

**Exercise 2.2.4.** Consider the diagram

$$\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{3} \mathbb{Z} \xrightarrow{4} \mathbb{Z} \xrightarrow{5} \mathbb{Z} \xrightarrow{6} \cdots$$

in the category $\mathsf{Ab}$ of abelian groups. It turns out the colimit of this diagram is $\mathbb{Q}$. How to show this?

For what follows, it will be useful to think of the diagram in (2.2.2) as a *functor* from the category

$$\mathcal{D} = \left\{ \begin{tikzcd} 0 \arrow[r] \arrow[d] & 1 \\ 2 & \end{tikzcd} \right\} \tag{2.2.5}$$

into $\mathsf{Grp}$. In this context, we might refer to (2.2.2) as a "$\mathcal{D}$*-shaped diagram in* $\mathsf{Grp}$."

**Definition 2.2.6.** Let $F : \mathcal{D} \to \mathcal{C}$ be a $\mathcal{D}$-shaped diagram in $\mathcal{C}$ (i.e., $F \in \mathsf{Fun}(\mathcal{D}, \mathcal{C})$). The *colimit functor* is the functor

$$\operatorname{colim} : \mathsf{Fun}(\mathcal{D}, \mathcal{C}) \to \mathcal{C}$$

sending $F$ to the object $\operatorname{colim} F$ in $\mathcal{C}$.

**Remark 2.2.7.** In general, *any* functor $F\,\mathcal{C} \to \mathcal{C}'$ can be thought of as a "$\mathcal{C}$-shaped diagram in $\mathcal{C}'$," and so we can always think about taking the colimit (or limit) of $F$. (Of course, it may often turn out that the colimit does not exist.)

**Remark 2.2.8.** It's worth remarking that colim is in fact a functor — i.e., a map $\eta : F \to F'$ of $\mathcal{D}$-shaped diagrams in $\mathcal{C}$ induces a map

$$\operatorname{colim} F \xrightarrow{\operatorname{colim} \eta} \operatorname{colim} F'.$$

**Example 2.2.9.** Here, a map of $\mathcal{D}$-shaped diagrams is a natural transformation of functors. For example, if $\mathcal{D}$ is the category in (2.2.5), and $F$ and $F'$ are the diagrams

$$F = \left\{ \begin{tikzcd} a \arrow[r] \arrow[d] & c \\ b & \end{tikzcd} \right\}, \qquad F' = \left\{ \begin{tikzcd} a' \arrow[r] \arrow[d] & c' \\ b' & \end{tikzcd} \right\}$$

in $\mathcal{C}$, then a morphism $\eta : F \to F'$ corresponds to a commutative diagram

$$\begin{tikzcd} a \arrow[r] \arrow[d] \arrow[dr, dashed, bend right, "\eta(0)"'] & b \arrow[r, dashed, bend left, "\eta(1)"] & a' \arrow[r] \arrow[d] & b' \\ c \arrow[rr, dashed, bend right, "\eta(2)"'] & & c' & \end{tikzcd}$$

But what *is* $\operatorname{colim} F$? Here we run into a problem: in general, we can't directly construct $\operatorname{colim} F$ like we did in Example 2.2.1, since in general we don't have any idea what the objects of $\mathcal{C}$ might be like (for example, they might not have "underlying sets"). The solution is to characterize colim via universal property.

**Construction 2.2.10** (Universal property of the colimit)**.** Fix a diagram category $\mathcal{D}$. (For illustrative purposes, let's choose $\mathcal{D}$ to be as in (2.2.5).) Let

$$\mathsf{const} : \mathcal{C} \to \mathsf{Fun}(\mathcal{D}, \mathcal{C})$$

be the "constant diagram functor" sending an object $x$ in $\mathcal{C}$ to the diagram

$$\mathsf{const}(x) = \left\{ \begin{tikzcd} x \arrow[r, "1_x"] \arrow[d, "1_x"'] & x \\ x & \end{tikzcd} \right\}.$$

Then we can *define* $\operatorname{colim} : \mathsf{Fun}(\mathcal{D}, \mathcal{C}) \to \mathcal{C}$ to be the left adjoint to $\mathsf{const}$. That is, we have

$$\operatorname{Hom}_{\mathcal{C}}(\operatorname{colim} F, x) = \operatorname{Hom}_{\mathsf{Fun}(\mathcal{D},\mathcal{C})}(F, \mathsf{const}\, x).$$

This is exactly a universal property of colim in the sense of Definition 2.1.2.

If $F$ is the diagram

$$F = \left\{ \begin{tikzcd} a \arrow[r] \arrow[d] & c \\ b & \end{tikzcd} \right\},$$

then the above construction is saying that there should be a natural bijection between the set maps arrows

$$f : \operatorname{colim} F \to x$$

in $\mathcal{C}$ and natural transformations

$$\begin{tikzcd}
a \arrow[r] \arrow[d] \arrow[drr, dashed, bend right] & b \arrow[drr, dashed, bend left] & & \\
 & & x \arrow[r, "1_x"] \arrow[d, "1_x"'] & x \\
c \arrow[drr, dashed, bend right] & & x &
\end{tikzcd}$$

If we squint hard enough at the following diagram, it maybe becomes believable that this is the case:

$$\begin{tikzcd}
a \arrow[r] \arrow[d] \arrow[ddrr, dashed, bend left] & b \arrow[d] \arrow[ddrr, dashed, bend left] & & \\
c \arrow[r] \arrow[ddrr, dashed, bend right] & \operatorname{colim} F \arrow[drr, "f"] & & \\
 & & x \arrow[r, "1_x"] \arrow[d, "1_x"'] & x \\
 & & x &
\end{tikzcd}$$

**Remark 2.2.11.** The universal property of limits should be "formally dual" to that of colimits. For example, we might just say the limit of a diagram $F : \mathcal{D} \to \mathcal{C}$ can be obtained by taking the colimit of the "dual" diagram $F^{\mathsf{op}} : \mathcal{D}^{\mathsf{op}} \to \mathcal{C}^{\mathsf{op}}$. That's technically valid but not super helpful. An (only slightly) more useful definition is to take the functor lim to be the *right* adjoint to $\mathsf{const}$.

To summarize, we have a double adjunction

$$\operatorname{colim} \dashv \mathsf{const} \dashv \lim .$$

## 2.3. **Digression: Interactions between** colim**,** lim**, and** Hom

A consequence of the formulation of the limit and colimit functors we discussed in the previous section is the following (*very* useful) fact.

**Theorem 2.3.1.** *Let $\mathcal{C}$ be a locally small category (i.e., the $\operatorname{Hom}_{\mathcal{C}}(c, c')$ is a* set *for every pair of objects $c, c'$ in $\mathcal{C}$) and let $\mathcal{D}$ be a small diagram category ($\mathcal{D}$ is locally small and has a* set *of objects which we'll call $d_1, d_2, d_2, \ldots$). Let $F : \mathcal{D} \to \mathcal{C}$ be a $\mathcal{D}$-shaped diagram in $\mathcal{C}$. Then, if $\operatorname{colim} F$ exists in $\mathcal{C}$*

$$\operatorname{Hom}_{\mathcal{C}}(\operatorname{colim} F, x) \cong \lim \operatorname{Hom}_{\mathcal{C}}(F(d_j), x)$$

*for any $x$ in $\mathcal{C}$, and where the limit on the right is taken in $\mathsf{Set}$. Dually, if $\lim F$ exists in $\mathcal{C}$, then*

$$\operatorname{Hom}_{\mathcal{C}}(x, \lim F) \cong \lim \operatorname{Hom}_{\mathcal{C}}(x, F(d_j)).$$

**Remark 2.3.2.** If you want to be very, very fancy, the first result in the above theorem can be rephrased concisely by saying that there is a natural bijection of sets

$$y(x)(\operatorname{colim} F) \cong \lim(y(x) \circ F),$$

where $y : \mathcal{C} \hookrightarrow \mathcal{P}(\mathcal{C})$ is the Yoneda embedding (Corollary 1.2.5). (Dually,

$$u(x)(\lim F) \cong \lim(u(x) \circ F),$$

where $u : \mathcal{C} \hookrightarrow \mathsf{Fun}(\mathcal{C}, \mathsf{Set})$ is the "covariant Yoneda embedding" of Remark 1.2.3.)

In other words (after identifying $\operatorname{colim} F = \lim F^{\mathsf{op}}$ where $F^{\mathsf{op}} : \mathcal{D}^{\mathsf{op}} \to \mathcal{C}^{\mathsf{op}}$), the following diagram of functors are commutative:

$$\begin{tikzcd}
\mathsf{Fun}(\mathcal{D}, \mathcal{C}) \arrow[r, "\operatorname{colim}"] \arrow[d, "\simeq"'] & \mathcal{C} \arrow[d, squiggly, "\simeq"] \\
\mathsf{Fun}(\mathcal{D}^{\mathsf{op}}, \mathcal{C}^{\mathsf{op}}) \arrow[r, "\lim"] \arrow[d, "y(x)_*"'] & \mathcal{C}^{\mathsf{op}} \arrow[d, "y(x)"] \\
\mathcal{P}(\mathcal{D}) \arrow[r, "\lim"] & \mathsf{Set}
\end{tikzcd} \tag{2.3.3}$$

**Exercise 2.3.4.** Convince yourself that Theorem 2.3.1 is true using the stuff we talked about in this section. Maybe also tracing through whatever (2.3.3) is saying might be interesting, too (i.e., pick a functor $F$ in the top left of the diagram and follow all the possible paths to the bottom right).

Somewhere to look for something written about this is [nLa25, "hom-functor preserves limits"]

Of course, what we've talked about above is what is commonly called "general abstract nonsense" [Mac71, Page 108]. Sometimes that's fun to think about—occasionally general nonsense ends up being extremely useful, though. Below are just a few examples.

**Example 2.3.5.** The Cartesian product is a kind of limit in $\mathsf{Set}$. This means

$$\mathrm{Hom}_{\mathsf{Set}}(A, \prod_i B_i) \cong \prod_i \mathrm{Hom}_{\mathsf{Set}}(A, B_i).$$

**Example 2.3.6.** Taking a quotient is a kind of colimit in the category of groups (as we saw in Example 2.2.1). Thus, if $N \leq G$ is a normal subgroup and $H$ is any other group,

$$\mathrm{Hom}_{\mathsf{Grp}}(G/N, H) \cong \lim \left( \begin{array}{ccc} & & \mathrm{Hom}_{\mathsf{Grp}}(G, H) \\ & & \downarrow \\ * = \mathrm{Hom}_{\mathsf{Grp}}(1, H) & \longrightarrow & \mathrm{Hom}(N, H) \end{array} \right).$$

This is another rephrasing of the universal property of the quotient. Its kind of nice because it means that to understand maps out of $G/N$ (which might be hard), we only need to understand maps out of $G$ and $N$, and how they relate.

**Example 2.3.7** (The fundamental group)**.** The category $\mathsf{Top}_*$ of *pointed topological spaces* has objects which are topological spaces $X$ equipped with a choice of "basepoint" $x \in X$ and maps $f : X \to Y$ which are continuous and such that $f(x) = y$ (i.e., "basepoint preserving maps"). The coproduct of two pointed spaces $X$ and $Y$ in this category is the "wedge sum," $X \vee Y$ which is like gluing together two pointed spaces at their basepoints.

Let $S^1$ be the circle. It turns out theres a nice map $\mathsf{pinch} : S^1 \to S^1 \vee S^1$ (by "pinching $S^1$ in the middle"). So using the trick of pulling a colimit out of Hom as a limit, we have a map

$$\mathrm{Hom}_{\mathsf{Top}_*}(S^1, X) \times \mathrm{Hom}_{\mathsf{Top}_*}(S^1, X) \cong \mathrm{Hom}_{\mathsf{Top}_*}(S^1 \vee S^1, X) \xrightarrow{\mathsf{pinch}^*} \mathrm{Hom}_{\mathsf{Top}_*}(S^1, X). \tag{2.3.8}$$

If we mod out $\mathrm{Hom}_{\mathsf{Top}_*}(S^1, X)$ by the equivalence relation of homotopy, we get an object

$$\mathrm{Hom}_{\mathsf{Top}_*}(S^1, X)/\text{homotopy} = \pi_1(X).$$

It turns out that the map in (2.3.8) "descends" to this quotient, and we get a map $\pi_1(X) \times \pi_1(X) \to \pi_1(X)$ which turns out to be associative and unital (with respect to the homotopy class of the constant map $S^1 \to x \in X$) and admits inverses. That is, there is a group structure on $\pi_1(X)$—this is called the *fundamental group* of $X$. There's some discussion of this in [Mil21, Page 96 and Definition 41.4].

**Remark 2.3.9.** A similar map $S^n \to S^n \vee S^n$ exists for all higher dimensional spheres ($n > 0$), and so we can similarly define the "higher homotopy groups" of a pointed space, $\pi_n(X)$. These turn out to be very interesting to study, be *very* difficult to compute even for "easy" spaces like spheres.

## References

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